

7 The curve C has equation $y = \frac{x^2 + 3x + 1}{x + 1}$.

- (a) Find the equations of the asymptotes of C . [3]

[illegible]

- (b)** Find the coordinates of any stationary points on C . [3]

[illegible]

- (c) Sketch C [3]

- (d) Sketch the curve with equation $y = \frac{|x|^2 + |x| + 1}{|x| + 1}$. [2]

- (e) Find, in exact form, the set of values of x for which $\frac{|x|^2 + |x| + 1}{|x| + 1} < 3$. [3]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for writing. The lines are evenly spaced across the entire page, providing a guide for letter height and placement. There is no text or other markings on the page.

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Additional pages

If you use the following page to complete the answer to any question, the question number must be clearly shown.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.